

1 Additional Standard Results

Proposition 1. *Let $K \subset \mathbb{R}_{++}$ be compact and let $f, f^n : K \rightarrow \mathbb{R}$ be continuous with $f^n \rightarrow f$ uniformly on K . Suppose f is strictly concave on K with unique maximizer $x = \arg \max_K f$, and $x^n \in \arg \max_K f^n$. Then $x^n \rightarrow x$ as $n \rightarrow \infty$.*

Proof. The sequence $\{x^n\} \subset K$ is bounded, so every subsequence has a further subsequence $x^{n_j} \rightarrow \bar{x} \in K$. Uniform convergence gives $f(\bar{x}) = \lim_j f^{n_j}(x^{n_j}) = \lim_j \max_K f^{n_j} = \max_K f = f(x)$, so by uniqueness of the maximizer $\bar{x} = x$. Every subsequence thus clusters only at x , hence $x^n \rightarrow x$. $\square$

Fact 1. *Let $g : X \rightarrow \mathbb{R}_{++}$ be a continuous function, where $X \subseteq \mathbb{R}^n$ is an open convex set. Define the weighted supremum norm $\|\cdot\|_g$ of a real-valued function $f : X \rightarrow \mathbb{R}$ by*

$$\|f\|_g := \sup_{x \in X} \frac{|f(x)|}{g(x)}. \quad (1)$$

If $\lim_{n \rightarrow \infty} \|f_n - f^\|_g = 0$, f_n converges to f^* uniformly on compact sets.*

Proof. Let K be a compact subset of X . Given the continuity of g , there is a positive maximum M_K of g on K . Then

$$\sup_{x \in K} |f_n(x) - f^*(x)| \leq M_K \sup_{x \in K} \frac{|f_n(x) - f^*(x)|}{g(x)} \leq M_K \|f_n - f^*\|_g. \quad (2)$$

Thus $\lim_{n \rightarrow \infty} \|f_n - f^*\|_g = 0$ implies $f_n \rightarrow f^*$ uniformly on K . $\square$

Fact 2. *Let $\{f_n\}$ be a sequence of continuous functions defined on a subset of the real line and converging uniformly to a function f on compact sets. If $\{x_n\}$ is a convergent sequence of real numbers with limit x , then $f_n(x_n)$ converges to $f(x)$.*

Proof. Let $K = \{x\} \cup \{x_n : n \geq N_0\}$; since $x_n \rightarrow x$, K is a compact subset of the common domain, and f is continuous on K as a uniform-on-compacts limit of continuous functions. For $n \geq N_0$,

$$|f_n(x_n) - f(x)| \leq \underbrace{|f_n(x_n) - f(x_n)|}_{\leq \sup_{y \in K} |f_n(y) - f(y)|} + |f(x_n) - f(x)|.$$

The first term $\rightarrow 0$ by uniform convergence on K ; the second $\rightarrow 0$ by continuity of f at x with $x_n \rightarrow x$. Hence $f_n(x_n) \rightarrow f(x)$. $\square$

Claim 1. *If f is convex and $f < 0$ on $(0, \lambda)$, then $\frac{f(s)}{s}$ is increasing on $(0, \lambda)$.*

Proof. Let f be convex on $(0, \lambda)$ and $f < 0$ on $(0, \lambda)$. Let x_1 and x_2 be two points in $(0, \lambda)$. Choose $0 < \alpha < x_1$. Then, any point (in particular, x_1) in (α, x_2) can be written as $x_1 = t\alpha + (1 - t)x_2$ for some $0 < t < 1$.

Now, define a new function F on $[\alpha, x_2]$ as:

$$F(x) = f(x) - f(\alpha).$$

Since f is convex, $F(x)$ is also convex on $[\alpha, x_2]$. To see this, observe that:

$$F(t\alpha + (1-t)x_2) = f(t\alpha + (1-t)x_2) - f(\alpha) \leq tf(\alpha) + (1-t)f(x_2) - f(\alpha) = tF(\alpha) + (1-t)F(x_2).$$

Since $F(\alpha) = 0$, the inequality simplifies to $F(x_1) \leq (1-t)F(x_2)$. Because $x_1 - \alpha = (1-t)(x_2 - \alpha)$, this gives $\frac{F(x_1)}{x_1 - \alpha} \leq \frac{F(x_2)}{x_2 - \alpha}$, i.e. the chord slope of the convex F from α is non-decreasing. Letting $\alpha \downarrow 0$ and writing $f(0^+) := \lim_{s \downarrow 0} f(s) \in [-\infty, 0]$, the slope $\frac{f(s) - f(0^+)}{s}$ is non-decreasing; since $f(0^+) \leq 0$, so is $\frac{f(0^+)}{s}$, hence for $y_1 \leq y_2$,

$$\frac{F(y_1)}{y_1} \leq \frac{F(y_2)}{y_2}.$$

Now, using the definition of $F(x)$, we have:

$$\frac{f(y_1)}{y_1} = \frac{F(y_1)}{y_1} + \frac{f(\alpha)}{y_1}.$$

Similarly, for y_2 :

$$\frac{f(y_2)}{y_2} = \frac{F(y_2)}{y_2} + \frac{f(\alpha)}{y_2}.$$

Since $\frac{F(y_1)}{y_1} \leq \frac{F(y_2)}{y_2}$ and $f(\alpha) < 0$, we conclude that:

$$\frac{f(y_1)}{y_1} \leq \frac{f(y_2)}{y_2}.$$

Thus, $\frac{f(s)}{s}$ is increasing on $(0, \lambda)$, completing the proof. □